



Lab 09

PH142 Spring 2026

Announcements

- **Lab09:** due 4/10 at 11:59pm
- **Quiz08:** due 4/11 at 12:00pm

Coming up:

- Extra credit for Participation (see Ed #266): due 4/26 at 10:00pm
- Data project part 3: due 5/1 at 11:59pm



Week 11 Lecture Review

One Sample Proportion Test

Goal: To test if a population proportion equals a claimed value

Assumptions:

- SRS with independent observations
- Sampling distribution is normal ($np_0 \geq 10$ and $n(1 - p_0) \geq 10$)

Week 11 Lecture Review

One Sample Proportion Test

Hypotheses: $H_0 : p = p_0$
 $H_a : p \neq p_0$ (or $<$ or $>$)

Test Statistic: $z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$

Week 11 Lecture Review

One Sample Proportion Test

Large Sample Method: $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$

Plus Four Method: $\tilde{p} = \frac{\text{number of successes} + 2}{n + 4}$

$$\tilde{p} \pm z^* \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n + 4}}$$

Week 11 Lecture Review

Two Sample Proportion Test

Goal: To test if two population proportions are the same or different.

Assumptions:

- SRS from each independent population
- Sampling distribution is normal (observed successes and failures are > 10 for both samples)

Week 11 Lecture Review

Two Sample Proportion Test

Hypotheses: $H_0 : p_1 - p_2 = 0$
 $H_a : p_1 - p_2 \neq 0$ (or $<$ or $>$)

Test Statistic:
$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Week 11 Lecture Review

Two Sample Proportion Test

Large Sample Method: $(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$

Plus Four Method: $\tilde{p}_1 = \frac{\text{no. of successes in pop1} + 1}{n_1 + 2}$

$$\tilde{p}_2 = \frac{\text{no. of successes in pop2} + 1}{n_2 + 2}$$

$$(\tilde{p}_1 - \tilde{p}_2) \pm z^* \sqrt{\frac{\tilde{p}_1(1 - \tilde{p}_1)}{n_1 + 2} + \frac{\tilde{p}_2(1 - \tilde{p}_2)}{n_2 + 2}}$$



LAB 09 Walkthrough

Lab Submission

- Follow the directions on the LAB09 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**